

- put cornary reflector in front of the radar
- 8 virtual antennas using 4 RX x 2 TX, narrower beam better precision
- add report with plots for Saeed after eth bug fix
- Saeed will call me to come to the parking and grab the second radar

Operate both radars together, find a way to sync both radars

Connect internal CLK's between them

minimize the time delay <10 ps, precisouly time synchronized (different from CLK freq)

- talk to Saeed once you get it working, to find time delta

“Two way time transfer”? Many many techniques

If you add, avg the CLK time period, i counted what ever is there (delta_t1 and delta_t2) and take the avg, or fiffnerentiate to get TOF

- To find CLK offset

using two radars, at least get better precision, larger aperture since now it becomes the width between the radars

The idea of the two way time transfer, i send a signal, and i listen for your signal to finish when yours come, precision of like 10 ps, ton of non-idealities, can introduce a ton of error,

Delayfrom CLK, receiver,, very challenging people cant do this, 100ps maybe, but 10ps is too tough and this is what we're trying to do

one way to do this, say, u send a pulse, i stop, typically u send pseudorandom bit stream, multiple pulses, and i have your code, you have mine, “matched filter” (review this), , add delay to multiply, correlation, until the match, when u get large correlation, to get the exact delay

Sequence of pulses to then use matched filter to decode

there are complications, wnat narrow pulses, he moment u narrow pulse , but problem is it requires lager BW

The first idea, someone has already done this (sadly), that there are limitations, normally weak filter that u receive that you receive, filter, amplify, then you need to digitize, sadly needs really fast ADC

if you send FMCW, i sent FMCW to you, you to me, when you mix, receive single freq, where it goes tells you time difference, dont need crazy ADC

instead of logn pulses, just do it by FMCW signaling

Another biggest advantage, slope? If slope is high, for the same time difference, you get larger frequency difference → there is a GAIN in it!

What he wants me to try to do, as soon as possible, (not just FMCW time transfer), first do it on MATLAB, then we get second unit, then do it through the two radars,

- 1) Simulate FMCW on MATLAB → get a plot or two to show the program manager
 - a) there are caveats here, so FMCW is trying to find time difference and CLK offset, converge to frequency difference, if we have originally frequency difference (this is a source of error, fixed with SYNC, assume fully SYNC plots)
 - b) for GPS ns, for this ps

We won't recognize who's FMCW → we need to code it, say phase coded, you can also code that phase of the chirp (this is coded FMCW) → should be easy once you do the regular FMCW on MATLAB

With coded,

Check CDMA (code division multiple access)

We send signals, his signal may be weaker, I want to recognize the signal, I need code, then you can correlate

- 2) for coded FMCW, just phase mod, look at what mapping they use

Take Harjani analog class?

- 3) take CLK from FMCW and put it into DDS (review whether the 9914 needs external CLK)
- 4) for phase mod, use differential VCO between outputs is analog or you can do it with DSP software

Once fix bug, get second unit, but then simulate before doing hardware

For now Saeed's is the same as the previous paper from 10 years ago, for now just implement this, then we can add secondary things.

- For example, these guys didn't have code, we do

Try to have the sims by Monday to Tuesday